

DANIEL SHASHANK DESHMUKH

Full-Stack Developer | AI Systems Engineer

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SUMMARY

Computer Science graduate (Mumbai University, 2026) building full-stack and applied-AI systems end to end. Shipped a POS for a working café, a GST-invoicing e-commerce platform, a hybrid-retrieval legal research engine for Indian law, and a typed, PyPI-published scientific library with 414 tests at 92.8% coverage.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL

Frontend / Backend: React 19, Next.js (App Router), Tailwind CSS, Vite, Recharts | FastAPI, Node.js/Express, REST, WebSockets, JWT/NextAuth, Pydantic, Zod

Data / AI: PostgreSQL (Neon), Turso/libSQL, SQLite, Prisma, Pinecone | RAG, hybrid retrieval (BM25 + dense, RRF), cross-encoder reranking, LoRA/PEFT, Groq, Gemini, Cohere, NVIDIA NIM

Quality / DevOps: pytest, Vitest, Playwright, Hypothesis, GitHub Actions, Lighthouse CI, Docker, Vercel, ruff, mypy, Git

WORK EXPERIENCE

Freelance Full-Stack Developer | Shree Gurudev Plastics, Mumbai

Aug 2026 – Present

- Building a wholesale e-commerce and operations platform on Next.js 15, TypeScript, Prisma and Turso: 64 API routes, 31 tables, a 1,500+ product catalog and a 23-section admin panel.
- Implemented GST-compliant invoicing (CGST/SGST/IGST split, HSN codes, jsPDF invoices, thermal receipts) and atomic order processing with server-side price validation and race-safe stock control.
- Integrated Razorpay (test mode) and Google OAuth; added WhatsApp click-to-chat enquiries and festival broadcast templates; 173 Vitest tests and CI with a deploy gate.

Freelance Full-Stack Developer | Babuji Chaay Café, Mumbai

Jul 2024 – Mar 2026

- Built a POS/sales PWA for a working café (billing with split cash/UPI, refunds with stock restoration, wastage logging, daily menu reset, Excel sales exports); later migrated it from Supabase to Next.js 15 + Turso (17 API routes, 8 tables).
- Built a promotions engine (percentage, fixed, BOGO, date-scheduled) and an automated daily "Lucky Customer" free-bill feature requiring no manual intervention; JWT-secured admin with PIN gate.
- Owned requirements, architecture, development and deployment; added a health-check endpoint and CI running TypeScript checks and 8 live API integration tests.

Customer Service Associate | Sahakari Bhandar (Reliance Retail), Mumbai

May 2024 – Dec 2024

- Handled high-volume billing and transactions accurately during peak hours across 100+ daily customer interactions.

PROJECTS

PYDECAY | Radioactive Decay Library | Python, NumPy/SciPy, React 19, PyPI | github.com/DanielDeshmukh/pydecay | pypi.org/project/pydecay

- Published a fully typed (mypy --strict) library with 41 public functions and classes: analytical decay, Bateman/matrix-exponential chains, multi-nuclide inventories, dose rates and shielding on ICRP-107 data (1,498 records); 7 PyPI releases.
- Validated with 414 tests at 92.8% coverage: differential checks against radioactivedecay and a 47-nuclide IAEA fixture, golden values at 1e-9 to 1e-12 tolerance, and 18 Hypothesis property tests.
- Shipped 4-job GitHub Actions CI (Python 3.10–3.13 matrix, coverage gate), MkDocs docs and a Vercel-hosted React demo with 248 Vitest tests.

VAULT | Permission-Aware Enterprise RAG (In Development) | FastAPI, Next.js, Postgres, Pinecone, Cohere |

github.com/DanielDeshmukh/vault

- Enforces access control before retrieval: an access-level filter is applied inside the Pinecone query, so unauthorized chunks are never fetched or sent to the LLM; 4 access levels, admin approval gate, 18 endpoints.
- Built a 50-question golden set and 9-persona evaluation harness (83% keyword recall) and verified auth and admin restrictions with 10 live access-control probes.

HECTOR | Legal Research Engine for Indian Law | FastAPI, Next.js, Pinecone, Groq | github.com/DanielDeshmukh/Hector |

hector-nine.vercel.app

- Built a hybrid retrieval pipeline fusing a custom BM25 with 2048-dim Pinecone vectors via reciprocal rank fusion, plus cross-encoder reranking, over a 17,876-vector index behind 26 endpoints.
- Added claim-level verification against retrieved sources (unverified claims flagged) and a query-intelligence layer scoring 94.2% on a 30-question eval; mapped IPC to BNS with a 496-entry table covering 485/511 sections.
- Secured with JWT and bcrypt API keys, 60 req/min rate limiting, input sanitization and log redaction; 1,164 tests (59% API coverage) and CI.

ELLA | Medical Triage Prototype | Python, Pinecone, BM25, LangChain, Gradio | github.com/DanielDeshmukh/ella

- Built a 3-stage hybrid retrieval pipeline (Pinecone, BM25, cross-encoder) over 90,306 chunks from 10 medical textbooks; raised intent-classification accuracy from 76% to 96% on a 50-query benchmark.
- Published ella-sdk to PyPI (21 releases, 12 tests, 78% package coverage) with automated tag-to-PyPI-to-HuggingFace release pipeline. Demo/prototype only; no clinical validation.

PROTEUS | Resume Analysis Toolkit | Next.js 16, Groq/Gemini, Turso | proteus-phi.vercel.app | github.com/DanielDeshmukh/proteus

- Multi-stage pipeline (JD parser, resume parser, deterministic gap scorer, rewrite suggester, cover-letter generator) with parallel stages, NDJSON streaming, 3-tier model fallback and a 3-hourly model health check; NextAuth (4 sign-in methods), 23 Playwright tests.

GITHUB-PROFILE-SCORE | Embeddable Profile Scorer | TypeScript, Express, Redis | github.com/DanielDeshmukh/github-profile-score

- API with 12 SVG endpoints and a 5-dimension scoring model; circuit breaker, jittered retry, in-flight dedup, two-tier cache and 214 tests. Fixed null-result cache poisoning and a GitHub 422 retry storm.

Additional: THEMIS – 4-bit LoRA fine-tune of Mistral 7B on ~52k Indian-law Q&A examples (Unsloth, free T4); published as the themis-llm CLI package on PyPI.

EDUCATION & CERTIFICATIONS

Bachelor of Computer Science, Shree L.R. Tiwari Degree College, Mumbai University | May 2026 | CGPI 7.38/10

Certifications: Anthropic (Claude API, Claude Code, Agent Skills, Sub Agents, MCP; 8 certificates, 2026) | Harvard CS50 AI with Python, CS50 SQL | freeCodeCamp (JavaScript Algorithms, Front-End Libraries, Responsive Web Design)